

七段译码器设计

1、实验任务：

学习共阳数码管 7 段数码显示译码器的硬件设计。

2、实验原理：

7 段数码是纯组合电路。通常的小规模专用 IC，如 74 或 4000 系列的器件只能作十进制 BCD 码译码，然而数字系统中的数据处理和运算都是二进制的，所以输出表达都是十六进制的。为了满足十六进制数的译码显示，最方便的方法就是利用 Verilog/VHDL 译码程序在 FPGA 中来实现。所以首先要设计一段程序。该程序可按照 case 语句表述方法，再根据真值表写出程序。设输入的 4 位码为 A[3:0]，输出控制 7 段共阳数码管（图 1.1）的 7 位数据为 LED7S[6:0]。输出信号 LED7S 的 7 位分别接图 1.1 的共阳数码管的 7 个段，高位在左，低位在右。例如当 LED7S 输出为“1101101”时，数码管的 7 个段 g、f、e、d、c、b、a 分别接 0、0、1、0、0、1、0；接有低电平的段发亮，于是数码管显示“5”。这里没有考虑表示小数点的发光管，如果要考虑，需要增加段 h，然后将 LED7S 改为 8 位输出。图 1.1，共阳数码示意图

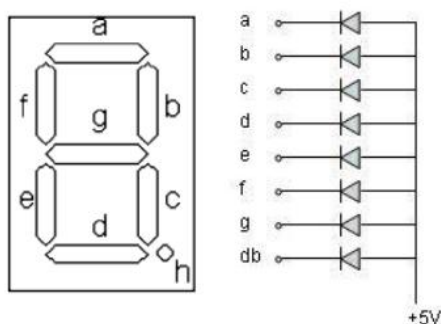


图 1.1 共阳数码管

3、实验程序：

```
module DECL7S (A,LED7S,B);
    input [3:0]  A;    //四位二进制输入
    output [6:0] LED7S; //动态数码管七段
    output reg[7:0] B;    //动态数码 8 位位选信号
    reg [6:0] LED7S;
    always @(A)
    begin
        B <= 8'b11111110;    //置最右侧数码管点亮，左侧七个数码管灭
    end
endmodule
```

```
case(A)
```

```

4'b0000 : LED7S <= 7'b1000_000;//7'b0111_111;//七段译码
4'b0001 : LED7S <= 7'b1111_001;//7'b0000_110;
4'b0010 : LED7S <= 7'b0100_100;//7'b1011_011;
4'b0011 : LED7S <= 7'b0110_000;//7'b1001_111;
4'b0100 : LED7S <= 7'b0011_001;//7'b1100_110;
4'b0101 : LED7S <= 7'b0010_010;//7'b1101_101;
4'b0110 : LED7S <= 7'b0000_010;//7'b1111_101;
4'b0111 : LED7S <= 7'b1111_000;//7'b0000_111;
4'b1000 : LED7S <= 7'b0000_000;//7'b1111_111;
4'b1001 : LED7S <= 7'b0010_000;//7'b1101_111;
4'b1010 : LED7S <= 7'b0001_000;//7'b1110_111;
4'b1011 : LED7S <= 7'b0000_011;//7'b1111_100;
4'b1100 : LED7S <= 7'b1000_110;//7'b0111_001;
4'b1101 : LED7S <= 7'b0100_001;//7'b1011_110;
4'b1110 : LED7S <= 7'b0000_110;//7'b1111_001;
4'b1111 : LED7S <= 7'b0001_110;//7'b1110_001;
default : LED7S <= 7'b1000_000;//7'b0111_111;

```

```
endcase
```

```
end
```

```
endmodule
```

4、实验仿真：

根据描述语言进行仿真测试，分析波形 图 1.2

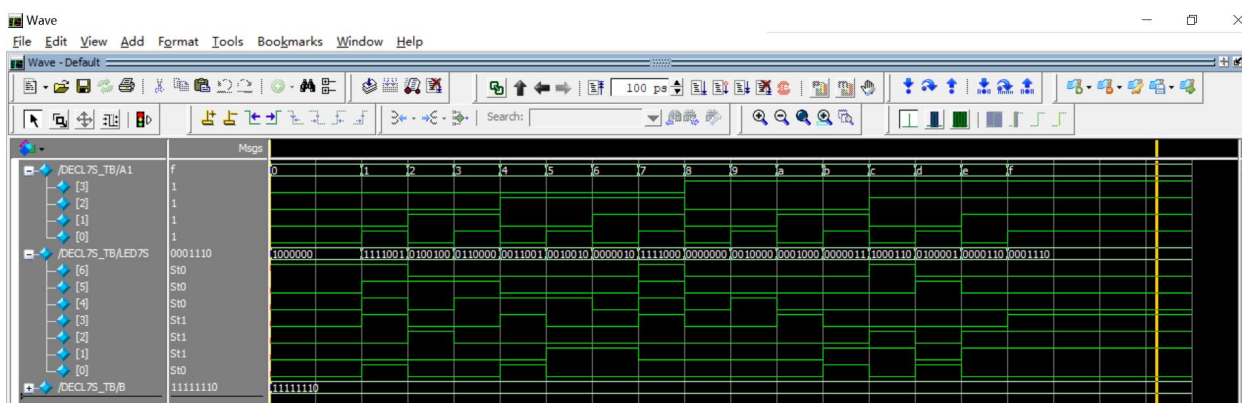
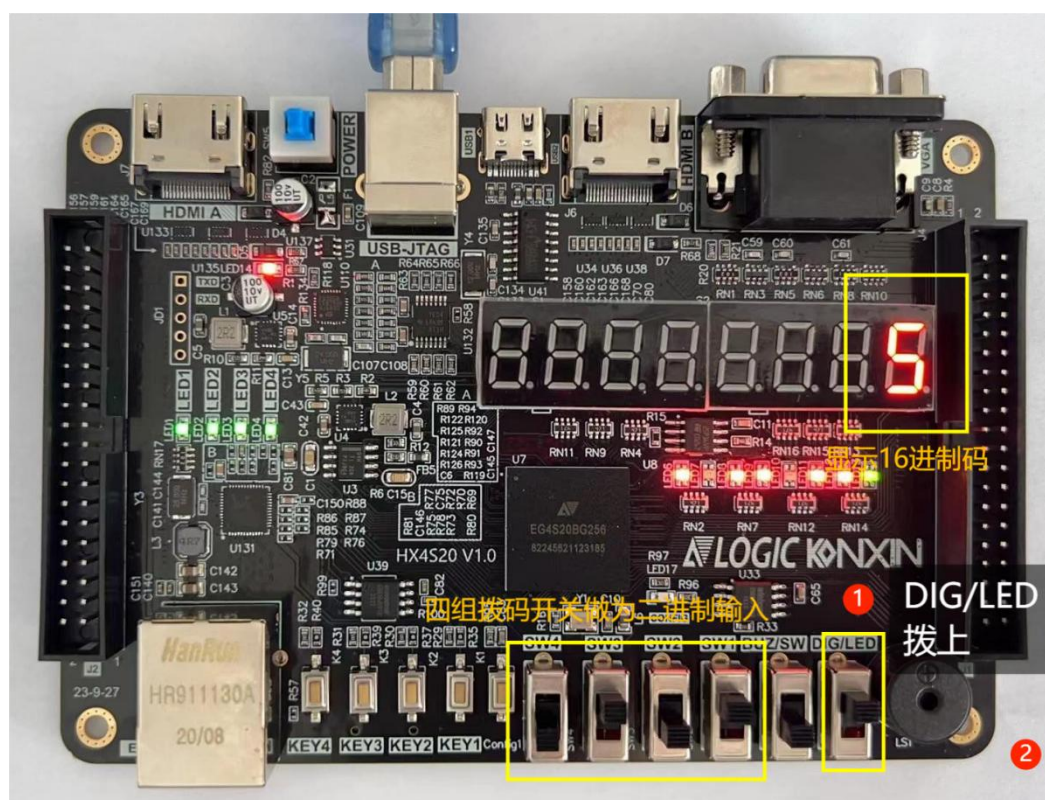


图 1.2 VWF 仿真结果

5、引脚约束：

```
set_pin_assignment { A[0] } { LOCATION = C8; IOSTANDARD = LVCMOS33; PULLTYPE =  
PULLUP; }  
set_pin_assignment { A[1] } { LOCATION = C7; IOSTANDARD = LVCMOS33; PULLTYPE =  
PULLUP; }  
set_pin_assignment { A[2] } { LOCATION = C6; IOSTANDARD = LVCMOS33; PULLTYPE =  
PULLUP; }  
set_pin_assignment { A[3] } { LOCATION = C5; IOSTANDARD = LVCMOS33; PULLTYPE =  
PULLUP; }  
set_pin_assignment { B[0] } { LOCATION = D11; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;  
PULLTYPE = NONE; }  
set_pin_assignment { B[1] } { LOCATION = E11; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;  
PULLTYPE = NONE; }  
set_pin_assignment { B[2] } { LOCATION = F10; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;  
PULLTYPE = NONE; }  
set_pin_assignment { B[3] } { LOCATION = C13; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;  
PULLTYPE = NONE; }  
set_pin_assignment { B[4] } { LOCATION = E10; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;  
PULLTYPE = NONE; }  
set_pin_assignment { B[5] } { LOCATION = E12; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;  
PULLTYPE = NONE; }  
set_pin_assignment { B[6] } { LOCATION = E16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;  
PULLTYPE = NONE; }  
set_pin_assignment { B[7] } { LOCATION = F16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;  
PULLTYPE = NONE; }  
set_pin_assignment { LED7S[0] } { LOCATION = C15; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { LED7S[1] } { LOCATION = C16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { LED7S[2] } { LOCATION = B6; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { LED7S[3] } { LOCATION = A5; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { LED7S[4] } { LOCATION = B5; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { LED7S[5] } { LOCATION = A8; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { LED7S[6] } { LOCATION = A7; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
```

6、下载测试:



SW1-SW4 分别输入 0000、0001、0010、0011...1111,看数码 1 效果。